



Kinetic coupling effects on the extinction limits of diffusion flames of hydrocarbons blended with ammonia

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ABSTRACT

Chemical kinetic coupling effects between NH_3 and liquid hydrocarbon fuels on extinction limits of diffusion flames are experimentally and numerically investigated. Three n-alkanes (n-heptane, n-decane, and n-dodecane), isooctane as a representative fully branched isoalkane, and toluene as a representative mono-aromatic are each tested, along with their blends with NH_3 . The measured extinction strain rates are analyzed, employing transport-weighted enthalpy and radical index to demonstrate relative changes of chemical kinetic potentials for each fuel evaluated. The results show a significantly lower chemical kinetic potential for NH_3 , compared to hydrocarbon fuels. Comparison of extinction limits as a function of transport-weighted enthalpy multiplied by radical index for fuel/ NH_3 mixtures shows potential promotive effects for n-alkanes and no significant coupling for isooctane and toluene. Planar laser-induced fluorescence is applied to quantify OH concentrations for n-heptane, isooctane, and their mixtures with NH_3 , and data are modeled to test the fidelity of chemical kinetic model predictions. It is found that including interaction reactions of n-alkyl and isoalkyl fragments with NH_2 , and the reactions involving methylamine and cyanide are critical to predicting OH production rates, as well as extinction limits for hydrocarbon/ NH_3 blends. Promotive effects of NH_3 blending on n-alkanes diffusion flame extinction limits are primarily from higher flame temperatures due to the reduced fraction of CO_2 found in the flame products. In the case of isooctane blended with NH_3 , the formation of two main isoalkyl fragments, CH_3 and C_3H_6 , are found to interact with NH_2 , resulting in the suppression of the reactive radical pool population and kinetic inhibition. A significantly weaker reactive radical pool in toluene oxidation leads to more significant inhibitive kinetic coupling from NH_3 -related reactions.

1. Introduction

Recently, ammonia (NH_3) has received substantial attention as a potential carbon-free energy carrier. NH_3 is known as an efficient hydrogen carrier with higher hydrogen contents per carrier atom than other carbon-based fuels such as methanol, ethanol, or gasoline [1–4]. However, the use of NH_3 in combustion applications imposes technical challenges not only because of its relatively lower energy density, but from its low reactivity, narrow flammability range, and slow burning velocity compared to hydrocarbon fuels. To overcome these challenges, dual fuel approaches (blending with hydrocarbon fuels) have been investigated [5–9]. The optimum blending ratio between diesel and NH_3 for diesel applications has been investigated to enhance ignition and burnout rate, also reductions in greenhouse gas and unburned NH_3 emissions [5,7–11]. Blending was restricted to <40 % energy fraction of

NH_3 to also constrain NO_x emissions. To characterize NH_3 blending with other fuels (e.g. hydrogen [12–15], methane [16,17], and dimethyl ether [18–20]), the ignition delay time, laminar burning velocity, and extinction behaviors were experimentally investigated. Adding NH_3 to large hydrocarbon fuels results in a non-linear decrease in laminar burning velocity [21].

While the impacts of NH_3 blending with light hydrocarbons have been extensively investigated, NH_3 blending with higher molecular weight (liquid) hydrocarbons relevant to transportation remain less examined, especially in terms of chemical kinetic coupling details. Recently, chemical kinetic model predictions and experiments in a high-pressure flow reactor and a supercritical jet-stirred reactor were compared [22], along with ignition delay time measurements in a rapid compression machine [23]. The inclusion of NH_2 reactions with n-heptane and n-heptane fragments (e.g., 1-butene, propene) were found to

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play important roles for predicting multi-stage ignition behaviors of n-heptane/ NH_3 mixtures. Pan et al. [24] investigated the chemical kinetic coupling between n-heptane and NH_3 oxidation in a jet-stirred reactor experiment, suggesting the importance of NO_x coupling with HO_2 and H radicals at intermediate temperature conditions.

The objective of this study is to investigate the chemical kinetic coupling of NH_3 with liquid hydrocarbon fuels on extinction behaviors of diffusion flames in a counterflow configuration. Considering the diverse molecular classes found in transportation fuels [25–27], three n-alkanes (n-heptane, n-decane, n-dodecane), isooctane, and toluene are blended with NH_3 to investigate NH_3 blending impacts. The measured extinction behaviors are analyzed using transport-weighted enthalpy and radical index parameters [28] to evaluate the potential chemical kinetic coupling between NH_3 , hydrocarbon fuels, and their reactive fragments. To assist in analyzing the kinetic coupling, three existing chemical kinetic models for NH_3 oxidation [22,29,30] are evaluated and integrated with an in-house chemical kinetic model for liquid hydrocarbon fuels [31]. Planar laser-induced fluorescence (OH-PLIF) imaging is applied to determine changes in OH concentrations with NH_3 addition to hydrocarbon fuel. After incorporating LIF quenching corrections, the observations are compared with modeling results. Chemical kinetic coupling impacts specific to fuel molecular classes for n-alkanes, a fully branched isoalkane, and a mono-aromatic are further elucidated using feature sensitivity and path flux analyses.

2. Experimental and numerical approaches

A counterflow burner (Fig. 1) is utilized to measure the extinction limits of pre-vaporized liquid hydrocarbon fuels and their mixtures with NH_3 . For the fuel side, the bottom burner (20 mm inner diameter at the nozzle exit) is as described in [32], where the fuel injector is relocated within a vaporization chamber. In the vaporization chamber, a liquid fuel is injected through a central nozzle along with nitrogen co-flow heated by electric heater, PID-controlled by monitoring the temperature at the outlet of the heater. The pre-vaporized fuel/nitrogen mixture is introduced at the bottom burner along with gaseous NH_3 . The vaporization chamber and burner are heated using band heaters, PID-controlled by monitoring flow temperatures at the exit of the chamber and burner surface, respectively. For consistent comparison to previous results [28], the temperature for the fuel-side at nozzle exit is maintained at 500 K. For oxidizer side, the upper burner (20 mm inner diameter nozzle exit) temperature is maintained at 300 K. The separation distance (L) between the nozzle exits is maintained at 15 mm.

Liquid fuel is controlled using a syringe pump (Harvard Apparatus, PHD 2000), while the flow rates of air, NH_3 , and nitrogen are determined using sonic nozzle systems and a mass flow controller (nitrogen co-flow), calibrated using a DryCal 800 (Mesa Labs). For liquid fuels, n-heptane (nC7), n-decane (nC10), n-dodecane (nC12), isooctane (iC8), and toluene are used to determine the extinction strain rates of diffusion flames with various fuel mole fractions (fuel/nitrogen mixtures). Blends of 25 % hydrocarbon fuel and 75 % NH_3 (mole fraction) are used to investigate the chemical kinetic coupling between hydrocarbon fuels

and NH_3 oxidations. The definition of global strain rate is used for extinction limits [28].

PLIF imaging is employed to quantify OH radical concentrations. A frequency-doubled Nd:YAG laser (Spectra-Physics, Quanta-Ray Pro-250-10) pumps a Dye laser (Spectra-Physics, Cobra-Stretch) with Rhodamine 590 to generate the excitation frequency for the $\text{Q}_1(6)$ OH transition (~ 283 nm). Cylindrical and plano-convex lenses are used to create a laser sheet beam (~ 150 μm thick, > 10 cm vertical height). Then, razor blades are used to make the laser beam into ~ 12 mm tall across the center of the burner to achieve relatively constant spatial laser power distribution. LIF signals are acquired in linear regime, requiring LIF quenching correction. The images are obtained using an Intensified CMOS camera (Andor, iStar sCOMS) with a 100 mm $f/2.8$ UV lens (Cercor) and two bandpass filters (UG-11 and WG-305).

Numerical calculations are performed using a counterflow diffusion flame configuration model in Cantera [33]. The two-point control continuation method is utilized to obtain extinction strain rates. The predictions of extinction limits by multiple chemical kinetic models are compared with the experimental results to evaluate model prediction fidelities. The model of Thorsen et al. [22] is reduced using path flux analysis [34], resulting in a total of 368 species to describe nC7/ NH_3 chemistries. Reduced RealFuel3 (RF3) models [31] for nC7, iC8, and toluene are used to compare with experimental measurements. Three different NH_3 models [22,29,30] are evaluated regarding their prediction fidelities for extinction limits. NH_3 and N_2 amine subsets from Xu et al. [30] are integrated into the reduced RF3 model (referred Model 1) [31] to keep the core chemistry from RF3. Finally, methylamine/cyanide subsets and hydrocarbon/ NH_2 abstraction reactions from Thorsen et al. [22] are added in Model 1 (referred Model 2).

3. Results and discussion

Fig. 2 summarizes the experimental extinction strain rates (a_E) for diffusion flames of pure nC7, nC10, nC12, iC8, toluene, NH_3 and hydrocarbon fuel/ NH_3 mixtures as a function of fuel mole fraction (X_f) (bottom-left). X_f is a fuel mole fraction in fuel/ N_2 mixture, where fuels are hydrocarbons, NH_3 , and their blends. Based upon X_f , nC12 exhibits the highest a_E among n-alkanes, followed by nC10 and nC7. The iC8 diffusion flames exhibit further reduced a_E , indicative of weaker chemical kinetic reactivity compared to those for n-alkanes. Toluene diffusion flames produce the lowest a_E among all the hydrocarbons tested, while NH_3 ($X_f = 1$) exhibits a significantly lower a_E still. NH_3 diffusion flames ($X_f < 1$) could not be examined due to their low reactivity. The measured a_E of hydrocarbon/ NH_3 mixtures reside between those of the individual pure hydrocarbons and that of NH_3 , indicating that NH_3 hinders mixture reactivity.

The extinction limits of diffusion flames of fuel/nitrogen mixtures in a counterflow configuration are primarily governed by three param-

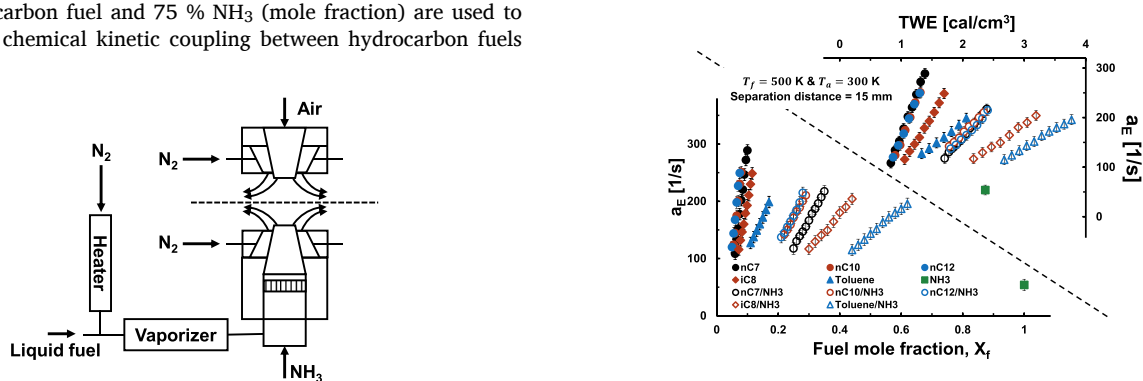


Fig. 1. Schematics of experiments.

Fig. 2. Measured extinction strain rates as a function of fuel mole fraction (bottom-left) and transport-weighted enthalpy (TWE) (top-right).

ters: 1) chemical enthalpy of fuel/nitrogen mixture; 2) diffusive transport characteristics of fuel molecules across the stagnation plane; and 3) chemical kinetic characteristics of the fuel [28]. To normalize the effects of chemical enthalpy and molecular diffusive transport, a “transport-weighted enthalpy” metric was developed as $TWE = [fuel] \times \Delta H_c \times (MW_{fuel}/MW_{nitrogen})^{-1/2}$ to compare the “chemical kinetic potential” for flames of different fuel mixtures. The top-right panel in Fig. 2 compares the measured a_E as a function of TWE. Note that the lower heating value (LHV) of each fuel is used for the heat of combustion (ΔH_c), while higher heating value (HHV) was used in [28]. While the use of either LHV or HHV doesn’t change overall trend of flame extinction behaviors, here LHV was used considering the exclusive formation of water in NH_3 combustion. The comparison in TWE value clearly demonstrates the commonality of chemical kinetic potential for all three n-alkanes diffusion flames.

The “chemical kinetic potential” of a fuel can be further elaborated upon by comparing the ability to populate the flame radical pool, referred to as the radical index (Ri) [28]. The Ri of all n-alkanes is defined as one, and those of iC8, toluene, and NH_3 then are found to be 0.7, 0.56, and 0.2, respectively through an examination of their model simulations. The Ri for fuel mixtures are calculated based on a linear mixing rule weighted by the mole fraction of each component, assuming no specific chemical kinetic coupling among the mixture components.

Fig. 3 compares the measured a_E as a function of $Ri \times TWE$. The dashed line represents the universal correlation developed in [28]. Except for n-alkanes/ NH_3 mixtures, all the measured a_E follow the universal correlation, suggesting no chemical kinetic coupling specific to iC8/ NH_3 and toluene/ NH_3 mixtures. A universal correlation appears regardless of using LHV or HHV in defining TWE. The a_E measurements for n-alkanes (nC7, nC10 and nC12) blended with NH_3 show higher values over the universal correlation, suggesting the potential promoting interaction of n-alkanes and NH_3 on a_E .

While the global extinction measurements suggest fuel-specific coupling between hydrocarbon fuels and NH_3 , it is important to compare the predictive fidelity for the various chemical kinetic models, particularly considering the level of current kinetic uncertainties. OH-PLIF imaging is employed to generate experimental validation data. The laser power was maintained at 8 ± 0.5 mJ/pulse. However, NH_3 diffusion flames produce more H_2O than, e.g., nC7 or iC8 flames. While H_2O is a major species quenching the LIF signal, NH_3 itself also contributes to quenching. The OH LIF quenching rate was theoretically evaluated using the collisional quenching rates proposed in [35]. Ten species (N_2 , H_2O , CO_2 , O_2 , CO , H_2 , N_2O , NO , CH_4 , and NH_3) were considered in calculating the fluorescence quantum yield, with species concentrations obtained from numerical calculations. The quenching effect on OH-PLIF imaging was experimentally verified by varying the delay time in 1 ns step at a fixed exposure time of 2 ns. Fig. 4 shows the decay of maximum OH intensity for nC7 and nC7/ NH_3 at $TWE = 1.98$ cal/cm³. Considering the laser pulse duration (~ 8 ns), measurements

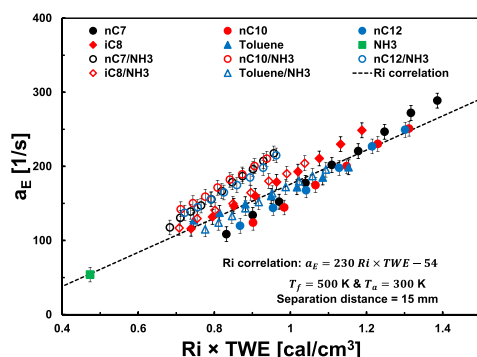


Fig. 3. Measured extinction strain rates as a function of radical index multiplied with transport-weighted enthalpy ($Ri \times TWE$).

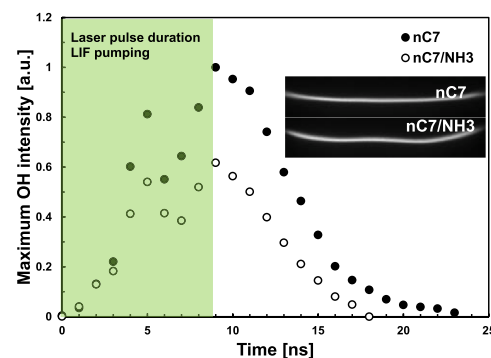


Fig. 4. Maximum OH LIF intensity for nC7 and nC7/ NH_3 as a function of laser delay time at $TWE = 1.98$ cal/cm³. Inserts are PLIF images for nC7 and nC7/ NH_3 mixture.

under 8 ns can be regarded as within the LIF pumping period. The OH signal decay time for nC7/ NH_3 mixture is ~ 22 % shorter than for pure nC7, in good agreement with the theoretical prediction.

Fig. 5 compares experimental results from OH-PLIF measurements with model predictions as a function of Damköhler number (Da), defined as $Da = a_E/a$. To clearly show the change of radical pool production rate dependency on NH_3 blending with either nC7 or iC8, the maximum OH concentration ($[OH]_{max}$) multiplied with strain rate (a), then normalized by Ri of each fuel is compared, as suggested in [28]. OH-PLIF measurements were performed at a constant TWE; $TWE = 1.98$ cal/cm³ for nC7 and nC7/ NH_3 flames, and 2.47 cal/cm³ for iC8 and iC8/ NH_3 flames. The peak temperatures of the tested flames vary from 1650 K to 1800 K, where $Q_1(6)$ exhibits constant temperature dependency. Considering the reasonable prediction on nC7 extinction behaviors by the model, the LIF results are calibrated against the model prediction for nC7 to yield semi-quantitative comparison. The experimental results show that nC7/ NH_3 mixture has higher OH production rates than pure nC7, while iC8/ NH_3 mixture exhibits similar or slightly lower OH production rates than iC8. While temperature (Boltzmann fraction) effects could be important for quantitative evaluation in LIF measurements [36], note that the measurements clearly show the enhanced OH production rate for nC7/ NH_3 mixtures over nC7.

For nC7/ NH_3 and iC8/ NH_3 mixtures, two models are compared to the experimental results. Model 1 does not include the nC7 fragment and NH_2 interaction reactions, whereas Model 2 incorporates the hydrocarbon + NH_2 abstraction reactions as well as the methylamine/cyanide chemistry subsets from Thorsen et al. [22]. Both models underpredict radical pool production rates for nC7/ NH_3 mixtures, more so for Model 2. However, Model 2 predicts an increased radical pool production rate because of NH_3 addition. iC8/ NH_3 mixture exhibits the inclusion of

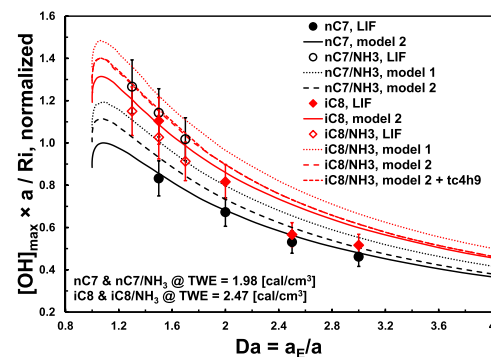


Fig. 5. Comparison of radical pool production rate between OH PLIF measurements and model predictions as a function of Damköhler number (Da). For nC7, only model 2 is shown, since both models exhibit identical behaviors.

methylamine/cyanide chemistry subsets and interaction reactions of Model 2 qualitatively improve emulation of the experimental results. This can be attributed to iC8 chemistry specific intermediate species. Considering that tC_4H_9 is one of the major intermediate species from iC8 oxidation, $\text{tC}_4\text{H}_9 + \text{NH}_2 = \text{tC}_4\text{H}_9\text{NH}_2$ and its subsequent oxidation reactions are added in Model 2 for the iC8/NH₃ mixture predictions to test the significance of C₄ amine chemistry. The high-pressure limit rate constant for $\text{tC}_4\text{H}_9 + \text{NH}_2 = \text{tC}_4\text{H}_9\text{NH}_2$ is estimated based on the geometric mean rule [37,38]. Subsequent reactions of $\text{tC}_4\text{H}_9\text{NH}_2$ (with rates by analogy to neopentane [39]) are assumed to form only $\text{iC}_4\text{H}_8\text{NH}_2$ then rapidly decompose to $\text{iC}_4\text{H}_8 + \text{NH}_2$, minimizing the number of additional species (See discussions in [30]). The model considering $\text{tC}_4\text{H}_9\text{NH}_2$ formation exhibits identical behaviors to Model 2, suggesting negligible contributions from $\text{tC}_4\text{H}_9\text{NH}_2$ reactions. Nevertheless, it is worth noting that treatment of C–N interactions in current kinetic models is far from comprehensive. For example, amine chemistry and unsaturated hydrocarbon + NH₂ addition pathways involving more than two carbon atoms are generally omitted. Furthermore, the present evaluation only considers one iC8 intermediate species (tC_4H_9). Further investigation of other key intermediate species from iC8 oxidation potentially interacting with NH₂ is warranted.

Fig. 6 compares the predicted a_E values from Model 1 and Model 2 against experimental results for nC7, iC8, toluene, NH₃, and fuel/NH₃ mixtures as a function of TWE. The reduced Thorsen et al. [22] model underpredicts nC7 measurements, while overpredicting NH₃ experimental observations. However, the Xu et al. [30] and Otomo et al. [29] model predictions are in reasonable agreements for the NH₃. While Model 1 captures overall trends, Model 2 predictions are more reasonably aligned with experiments for all three fuel/NH₃ mixtures. Fig. 7 shows that Model 2 also adequately reproduces experimental results for a_E as a function of $Ri \times TWE$ for all hydrocarbon fuels, predicting the increase in a_E for nC7/NH₃ mixture and indiscernible coupling for the iC8/NH₃ and toluene/NH₃ mixtures. Considering the experimental uncertainties, the comparison in Fig. 6 shows slight improvements in predictive fidelity, insufficient to support the updates in Model 2. By employing the $Ri \times TWE$ metric (Fig. 7), Model 2 more appropriately captures the promotive/inhibitive impacts specific to each type of fuel molecular structure.

To further elucidate chemical kinetic coupling between NH₃ and hydrocarbon fuels, sensitivity and path flux analyses are performed with nC7/NH₃, iC8/NH₃, and toluene/NH₃ mixtures using Model 2. Data are obtained at the extinction strain rate and constant $Ri \times TWE$ ($= 0.9 \text{ cal/cm}^3$). Although not shown in Fig. 8, the major reaction pathways for heat release are found to be $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$ and $\text{H}_2 + \text{OH} = \text{H} + \text{H}_2\text{O}$, whereas adding NH₃ to hydrocarbon fuels results in $\sim 9\%$ of total heat release through $\text{NH}_2 + \text{NO} = \text{H}_2\text{O} + \text{N}_2$. About 50 % of OH radical production results from $\text{H} + \text{O}_2 = \text{O} + \text{OH}$, which is highly temperature dependent.

Fig. 8a compares the OH consumption channels. Pure hydrocarbon

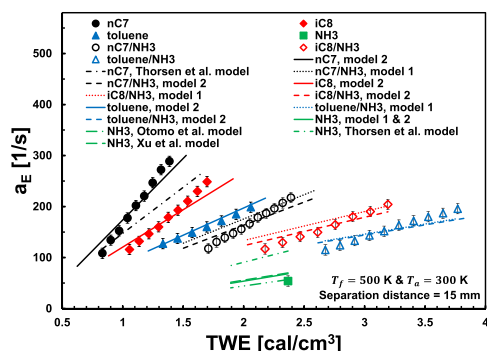


Fig. 6. Comparison of the predicted a_E values against measurements in TWE for nC7, iC8, toluene, NH₃, and their mixtures with NH₃.

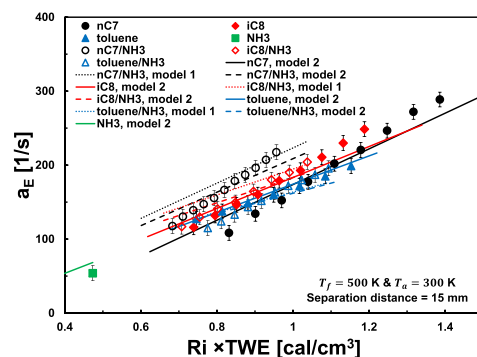


Fig. 7. Comparison of the predicted a_E values against measurements in $Ri \times TWE$ for nC7, iC8, toluene, NH₃, and their mixtures with NH₃.

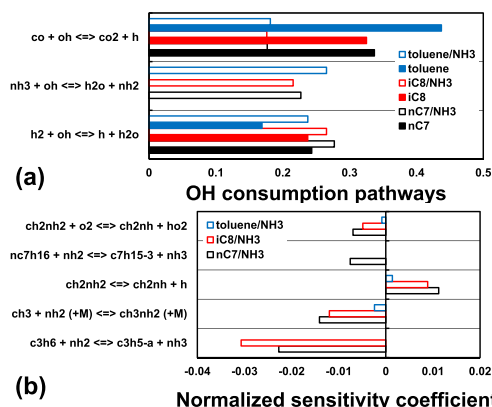


Fig. 8. (a). Comparison of OH consumption pathway among fuels tested. (b) Sensitivity analysis at extinction limits for interaction reactions for nC7/NH₃, iC8/NH₃, and toluene/NH₃ mixtures.

fuels consume OH mainly through $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$ and $\text{H}_2 + \text{OH} = \text{H} + \text{H}_2\text{O}$, which are dominant exothermic reactions. The addition of NH₃ results in considerable consumption of OH through $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$, consuming most of the NH₃ at near-extinction conditions. Fig. 8b shows the normalized sensitivity coefficients for key interaction reactions affecting the extinction limit for the fuel/NH₃ mixtures. Except for $\text{CH}_2\text{NH}_2 = \text{CH}_2\text{NH} + \text{H}$, the interaction reactions are inhibitive. The reactions $\text{C}_3\text{H}_6 + \text{NH}_2 = \text{C}_3\text{H}_5\text{-a} + \text{NH}_3$ and $\text{CH}_3 + \text{NH}_2 (+M) = \text{CH}_3\text{NH}_2 (+M)$ are the most inhibitive reactions for nC7/NH₃ and iC8/NH₃ mixtures. These reactions have no influence on toluene/NH₃, since none of the major n-alkyl and isoalkyl fragments involved in the alkane cases are produced.

Fig. 9a compares the mole fractions of CH₃ and C₃H₆; both species exhibit higher mole fractions in the iC8/NH₃ mixture, compared to the nC7/NH₃ mixture. Based on the CH₃ flux analysis, 11 % of CH₃ is consumed by $\text{CH}_3 + \text{NH}_2 (+M)$ in nC7/NH₃ mixture, 5 % of CH₃ is consumed by that reaction in iC8/NH₃ mixture. Even though the fraction of the reaction is lower in iC8/NH₃ mixture, the greater amount of CH₃ can be attributed to direct formation from iC8 fragments, resulting in the greater production of CH₃NH₂ compared to the nC7/NH₃ mixture. Based on the flux analysis of iC8 and its fragments, CH₃ is mainly produced by fuel decomposition, $\text{iC}_8\text{H}_{18} = \text{CH}_3 + \text{yC}_7\text{H}_{15}$ (53.2 %) and $\text{iC}_8\text{H}_{18} = \text{CH}_3 + \text{pC}_7\text{H}_{15}$ (8.7 %) and fuel fragment decomposition, $\text{iC}_4\text{H}_9 = \text{C}_3\text{H}_6 + \text{CH}_3$. Methylamine (CH_3NH_2) is a metastable species, inhibiting overall reactivity and lowering the extinction strain rate. C₃H₆ is produced principally by $\text{C}_3\text{H}_5\text{-a} + \text{H} (+M) = \text{C}_3\text{H}_6 (+M)$ in nC7/NH₃ mixtures and produced principally by $\text{iC}_3\text{H}_7 = \text{C}_3\text{H}_6 + \text{H}$ in iC8/NH₃ mixtures. isoPropyl radicals (iC_3H_7) are produced by iC8 fragment

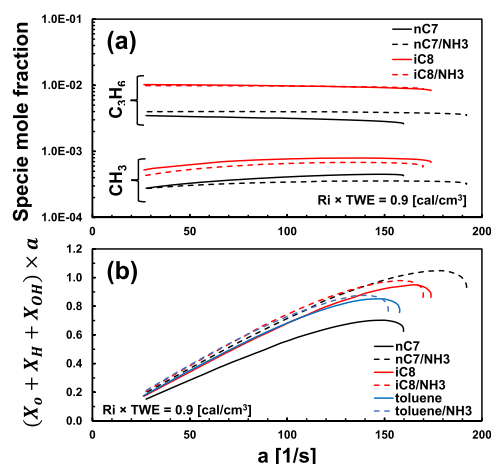


Fig. 9. (a). Comparison of key isoalkyl fragments, CH_3 and C_3H_6 as a function of strain rate for nC7, iC8, and their mixtures with NH_3 . (b) Radical pool production rates defined as the summation of three key radicals (O, H, and OH) multiplied with strain rate for nC7/ NH_3 , iC8/ NH_3 , and toluene/ NH_3 mixtures.

decomposition, $\text{yC}_7\text{H}_{15} = \text{iC}_3\text{H}_7 + \text{iC}_4\text{H}_8$. About 60 % of C_3H_6 produces C_3H_5 -a mostly through $\text{C}_3\text{H}_6 + \text{H} = \text{C}_3\text{H}_5\text{-a} + \text{H}_2$ (23 %) and $\text{C}_3\text{H}_6 + \text{NH}_2 = \text{C}_3\text{H}_5\text{-a} + \text{NH}_3$ (20 %), and 37 % of C_3H_5 -a produces C_3H_6 back through $\text{C}_3\text{H}_5\text{-a} + \text{H} (+\text{M}) = \text{C}_3\text{H}_6 (+\text{M})$. Therefore, having more C_3H_6 produced for iC8/ NH_3 mixture consumes more H to produce C_3H_6 and C_3H_5 -a in cycle, resulting in an inhibitive effect on extinction.

To demonstrate the thermal and/or kinetic impact of NH_3 blending on nC7, iC8, and toluene, the total (O, H, and OH) radical pool production rates are defined by the summation of the key radical concentrations multiplied by the strain rate in Fig. 9b. The radical pool populations for iC8 vs. iC8/ NH_3 and toluene vs. toluene/ NH_3 result in similar extinction behaviors, while those for the nC7/ NH_3 mixture show a higher radical pool population compared to that of nC7. The key reactive radicals involved in diffusion flame extinction are H, OH, and O, which are produced through the chain branching reactions as discussed above. At low strain rate far from extinction, fuel molecules are primarily consumed through pyrolysis reactions, i.e., thermal decomposition reactions and H abstraction reactions by H radicals [28]. Near the extinction limit, the OH radical emerges in terms of fuel oxidation reactions that yield water and compete with $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$, reducing the production rate of H atoms. The comparison at constant $Ri \times TWE$ inevitably results in a larger initial fuel mole fraction for fuel mixtures having a lower Ri value. On the other hand, the blending of NH_3 results in a higher flame temperature due to the reduced CO_2 concentration and lower specific heat of combustion products. Consequently, NH_3 blending at the constant $Ri \times TWE$ exhibits a higher mole fraction of OH radicals. n-Alkyl fragments produced in nC7/ NH_3 oxidation result in reducing the mole fractions of key species (CH_3 and C_3H_6) interacting with NH_2 . Consequently, nC7/ NH_3 mixtures produce higher values of a_E over those for pure nC7. CH_3 and C_3H_6 are primary intermediate species produced from isoalkyl fragments yielded in iC8/ NH_3 combustion, both kinetically interacting with NH_2 produced from NH_3 oxidation. These interactions inhibit the production of H and overall reactivity. Toluene competes for H (~52 % of total toluene consumption pathways) over H abstraction from NH_3 . Combined with the lower production rate of H atoms in comparison to alkane oxidations, the radical pool of the toluene flame is significantly weaker. While no NH_2 interactions with toluene-specific fragments are currently present in Model 2, the extinction behaviors of toluene/ NH_3 mixtures are reasonably well predicted for the constant $Ri \times TWE$ condition. In conclusion, all three fuel/ NH_3 mixtures produce more OH radicals due to the increased flame temperatures, however, iC8/ NH_3 and toluene/ NH_3 mixtures have inhibitive chemical kinetic impacts on H radicals,

resulting in no promotive effects on total radical pool populations.

4. Conclusions

The impact of NH_3 blending on extinction limits of diffusion flames in a counterflow configuration was experimentally and numerically investigated. Results for three n-alkanes (n-heptane, n-decane, and n-dodecane), a fully branched isoalkane (isooctane) and a mono-aromatic (toluene) were used to investigate the fragment effects yielded from the representative fuel structures and their interactions with NH_3 on extinction limit. Experimentally determined extinction strain rates were analyzed employing the “transport-weighted enthalpy” to normalize thermal and transport effects, and to compare relative behaviors due to differences in the “chemical kinetic potentials” of tested fuel mixtures. Comparisons of the transport-weighted enthalpy metric for each mixture demonstrated a significantly reduced chemical kinetic potential for NH_3 compared to the hydrocarbon fuels. Comparison of extinction limits in terms of the transport-weighted enthalpy multiplied by the “radical index”, assuming no specific chemical kinetic interactions of NH_3 , hydrocarbons, and/or their fragments, revealed a promoting effect of n-alkane observations by NH_3 addition, but no effect from addition to the isooctane and toluene. Kinetic coupling was further investigated experimentally by comparing OH concentrations for n-heptane, isooctane, and their mixtures with NH_3 . Planar laser-induced fluorescence for OH observations taken at constant transport-weighted enthalpy conditions were corrected for differences in LIF quenching rates. The corrected OH measurements were compared with chemical kinetic model predictions to elucidate observed changes in radical pool concentrations from blending NH_3 into the two hydrocarbon fuels.

Comparisons revealed the importance of interaction reactions of n-alkyl and isoalkyl fragments with NH_2 , and the reactions of methylamine and cyanide in terms of predicting semi-quantitatively observed OH production rates and extinction limits. Further chemical kinetic modeling at a constant product of transport-weighted enthalpy and radical index revealed that promotion of extinction limit by NH_3 blending primarily stemmed from reduced presence of CO_2 , as a reaction product and hence, higher flame temperatures. isoOctane oxidation generates principally CH_3 and C_3H_6 fragments that interact with NH_2 , yielded from NH_3 , suppress the H radical population, thus inhibitive kinetic coupling. The significantly weaker radical pool of toluene compared to alkane flames accentuated the inhibitive kinetic coupling resulting from toluene and NH_3 competing for H radicals.

Considering the diverse molecular structural composition of petroleum as well as emerging sustainable and/or renewable liquid fuels, the results in this work suggest structurally specific chemical kinetic coupling through co-firing of liquid hydrocarbons with NH_3 needs to be carefully considered in terms of controlling applied combustion behaviors and emissions in energy conversion systems.

Novelty and significance statement

The novelty of this study is that the coupling impacts between NH_3 and liquid hydrocarbon fuels on diffusion flame extinction are thoroughly evaluated by comprehensive analysis of experimental results, thus identifying the significance of fuel-specific interaction reactions between NH_3 and liquid hydrocarbon fuels. The results are particularly important for co-firing approaches of NH_3 and real liquid fuels, where the fuel specific chemical kinetic interaction reactions between NH_3 and liquid fuel influence global combustion behaviors.

Author contributions

- MKY performed research, analyzed data, drafted and authored the paper.
- FLD analyzed, edited and authored the paper.
- MPB analyzed kinetic interpretation and authored the paper.

- SHW designed research, analyzed data, drafted, and authored the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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